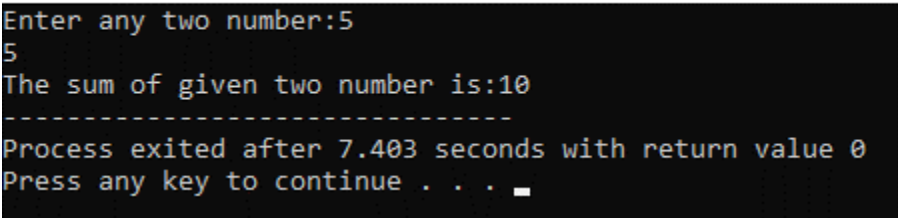


Lab-5 (Function)

1. Write a c-program to create a function named sum to calculate sum of two numbers.

Ans:

```
#include<stdio.h>
int sum(int a, int b);
int main()
{
    int result;
    int a,b;
    printf("Enter any two number:");
    scanf("%d%d",&a,&b);
    result=sum(a,b);
    printf("The sum of given two number is:%d",result);
    return 0;
}
int sum(int a,int b)
{
    int c;
    c=a+b;
    return c;
}
```

Output: 

2. Write a c-program to calculate area of a circle using concept of function.

Ans:

```
#include<stdio.h>
int area(int r);
int main()
{
    float a,r;
    printf("Enter radius of circle:");
    scanf("%f",&r);
    a=area(r);
    printf("The area of circle of is %f",a);
    return 0;
}
```

```

}
int area(int r)
{
    int c;
    c=3.1415*r*r;
    return c;
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\function.exe

```

Enter radius of circle:4
The area of circle of is 50.000000
-----
Process exited after 4.981 seconds with return value 0
Press any key to continue . . .

```

3. Write a c-program to create a function name prime to check whether entered number is prime or not.

Ans:

```

#include<stdio.h>
int prime(int a);
int main()
{
    int a,c;
    printf("Enter any number:");
    scanf("%d",&a);
    c=prime(a);
    if(c==1)
        printf("The number is prime");
    else
        printf("The number is not prime");
    return 0;
}
int prime(int a)
{
    int i,count=0;
    for(i=1; i<=a; i++)
    {
        if(a%i==0)
        {
            count=count+1;
        }
    }

    if(count==2)

```

```

    return 1;
    else
    return 0;
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\function.exe

```

Enter any number:13
The number is prime
-----
Process exited after 4.114 seconds with return value 0
Press any key to continue . . .

```

4. Write a c-program to find factorial of number using recursion function.

Ans:

```

#include<stdio.h>
int fact(int);
int main()
{
    int n,f;
    printf("Enter any number:");
    scanf("%d",&n);

    f=fact(n);
    if(n<0)
    printf("There is no factorial for negative number");
    else
    printf("The factorial of %d is:%d",n,f);
    return 0;
}
int fact(int n)
{
    if(n>1)
    {
    return n*fact(n-1);
    }
    else
    {
    return 1;
    }
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\function.exe

```

Enter any number:6
The factorial of 6 is:720
-----
Process exited after 2.49 seconds with return value 0
Press any key to continue . . .

```

5. Write a c-program find Fibonacci using recursion function.

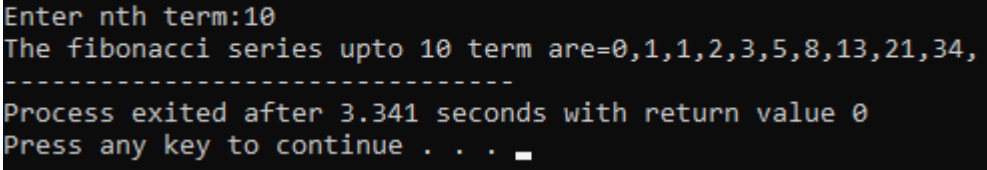
Ans:

```
#include<stdio.h>
int fibo(int);
int main()
{
    int i,n;
    printf("Enter nth term:");
    scanf("%d",&n);
    printf("The fibonacci series upto %d term are=",n);
    for(i=0; i<n; i++)
    {
        printf("%d,",fibo(i));
    }
    return 0;
}
int fibo(int n)
{
    if(n==0)
        return 0;

    else if (n == 1)
        return 1

    else
        return fibo(n-1)+fibo(n-2);
}
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\function.exe



```
Enter nth term:10
The fibonacci series upto 10 term are=0,1,1,2,3,5,8,13,21,34,
-----
Process exited after 3.341 seconds with return value 0
Press any key to continue . . .
```

6. Write a c-program to find sum of two matrix input by user using function.

Ans:

```
#include<stdio.h>
void sum(int row, int col, int a[row][col], int b[row][col]);
int main()
{
    int row,col,i,j;
    printf("Enter row and column");
    scanf("%d%d",&row,&col);
```

```

int a[row][col],b[row][col];
printf("Enter an element for first matrix");
for(i=0; i<row; i++)
{
    for(j=0; j<col; j++)
    {
        scanf("%d",&a[i][j]);
    }
}

printf("Enter an element for second matrix");
for(i=0; i<row; i++)
{
    for(j=0; j<col; j++)
    {
        scanf("%d",&b[i][j]);
    }
}
sum(row,col,a,b);
}

void sum(int row, int col, int a[row][col], int b[row][col])
{
    int i,j,s[row][col];
    for(i=0; i<row; i++)
    {
        for(j=0; j<col; j++)
        {
            s[i][j]=a[i][j]+b[i][j];
        }
    }

    printf("The sum of matrix is:\n");
    for(i=0; i<row; i++)
    {
        for(j=0; j<col; j++)
        {
            printf("%d\t",s[i][j]);
        }
        printf("\n");
    }
}

```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\function.exe

```
Enter row and column2
3
Enter an element for first matrix1 2 3 4 5 6
Enter an element for second matrix1 2 3 4 5 6
The sum of matrix is:
2      4      6
8      10     12

-----
Process exited after 38.5 seconds with return value 2
Press any key to continue . . .
```

7. Write a c-program to swap two number input by user using call by value method.

Ans:

```
#include<stdio.h>
void swap(int ,int );
int main()
{
    int a,b;
    printf("Enter a first number");
    scanf("%d",&a);
    printf("Enter a second number");
    scanf("%d",&b);
    swap(a,b);
    return 0;
}
void swap(int a, int b)
{
    a=a+b;
    b=a-b;
    a=a-b;
    printf("The value after swap is:%d,%d",a,b);
}
```

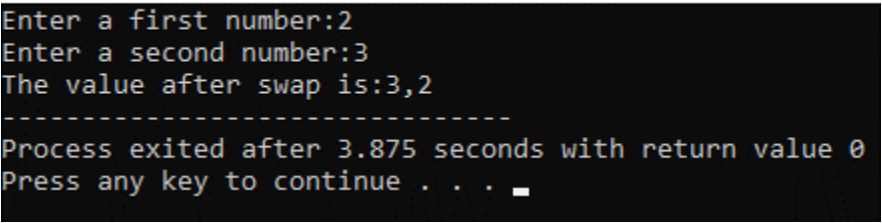
Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\function.exe

```
Enter a first number2
Enter a second number3
The value after swap is:3,2
-----
Process exited after 3.63 seconds with return value 0
Press any key to continue . . .
```

8. wap to swap two number input by user using call by reference method.

Ans:

```
#include<stdio.h>
void swap (int *x, int *y);
int main()
{
    int a,b;
    int *x;
    int *y;
    x=&a;
    y=&b;
    printf("Enter a first number:");
    scanf("%d",&a);
    printf("Enter a second number:");
    scanf("%d",&b);
    swap(x,y);
    return 0;
}
void swap(int *x, int *y)
{
    *x=*x+*y;
    *y=*x-*y;
    *x=*x-*y;
    printf("The value after swap is:%d,%d",*x,*y);
}
```

Output: 

9. Write a c-program to reverse a number input by user using function.

Ans:

```
#include<stdio.h>
int reverse(int num);
int main()
{
    int num,rev;
    printf("Enter any number:");
    scanf("%d",&num);
    rev=reverse(num);
}
```

```

        printf("The number in reverse is: %d",rev);
    }
    int reverse(int num)
    {

    int rem,result=0;
        while(num!=0)
        {
            rem=num%10;
            result=result*10+rem;
            num=num/10;
        }
        return result;
    }

```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\function.exe

```

Enter any number:12345
The number in reverse is: 54321
-----
Process exited after 5.904 seconds with return value 31
Press any key to continue . . .

```

10. Write a c-program to input element of array and display it using function.

Ans:

```

#include<stdio.h>
void array(int a[],int size);
int main()
{
    int size,i;
    printf("Enter size of array:");
    scanf("%d",&size);
    int a[size];
    printf("Enter an element:");
    for(i=0; i<size; i++)
    {
        scanf("%d",&a[i]);
    }
    array(a,size);
    return 0;
}
void array(int a[], int size)
{
    int i;
    printf("The element of array are:");
    for(i=0; i<size; i++)

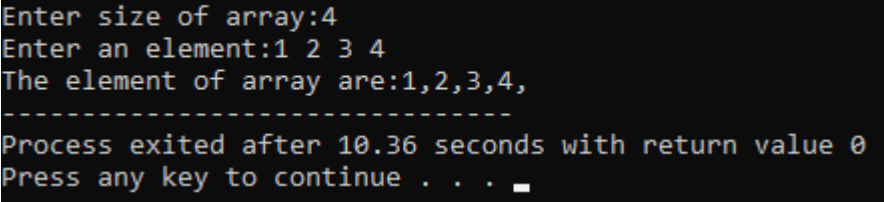
```

```

    {
        printf("%d",a[i]);
    }

}

```

Output: 

11. Write a c-program to find maximum element of 1d array using function.

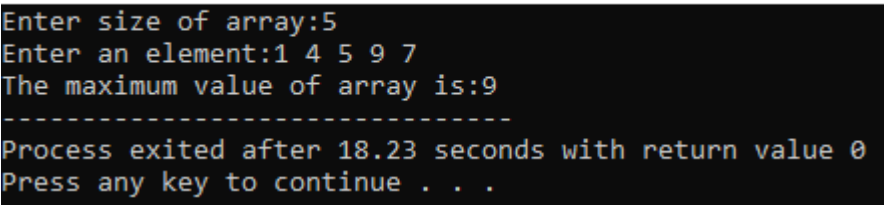
Ans:

```

#include<stdio.h>
int maximum(int a[],int size);
int main()
{
    int size,i,max;
    printf("Enter size of array:");
    scanf("%d",&size);
    int a[size];
    printf("Enter an element:");
    for(i=0; i<size; i++)
    {
        scanf("%d",&a[i]);
    }
    max=maximum(a,size);
    printf("The maximum value of array is:%d",max);
    return 0;
}
int maximum(int a[],int size)
{
    int i,max;
    max=a[0];
    for(i=0; i<size; i++)
    {
        if(max<a[i])
        {
            max=a[i];
        }
    }
    return max;
}

```

```
}
```

Output:  C:\Users\Aayus\OneDrive\Desktop\New folder\C-program\function.exe

```
Enter size of array:5
Enter an element:1 4 5 9 7
The maximum value of array is:9
-----
Process exited after 18.23 seconds with return value 0
Press any key to continue . . .
```

12. Write a c-program to input matrix and find the transpose using function.

Ans:

```
#include<stdio.h>
```

```
void transpose(int row,int col,int a[row][col]);
```

```
int main()
```

```
{
```

```
    int row,col,i,j;
```

```
    printf("Enter size of row and column");
```

```
    scanf("%d%d",&row,&col);
```

```
    int a[row][col];
```

```
    printf("Enter an element:");
```

```
    for(i=0; i<row; i++)
```

```
    {
```

```
        for(j=0; j<col; j++)
```

```
        {
```

```
            scanf("%d",&a[i][j]);
```

```
        }
```

```
    }
```

```
    transpose(row,col,a);
```

```
    return 0;
```

```
}
```

```
void transpose(int row,int col,int a[row][col])
```

```
{
```

```
    int b[col][row];
```

```
    int i,j;
```

```
    for(i=0; i<col; i++)
```

```
    {
```

```
        for(j=0; j<row; j++)
```

```
        {
```

```
            b[i][j]=a[j][i];
```

```
        }
```

```
    }
```

```
    printf("The transpose of matrix is:\n");
```

```

    for(i=0; i<col; i++)
    {
        for(j=0; j<row; j++)
        {

            printf("%d\t",b[i][j]);

        }
        printf("\n");
    }
}

```

Output:  C:\Users\aaayus\OneDrive\Desktop\New folder\C-program\function.exe

```

Enter size of row and column2 2
Enter an element:1 2 3 4
The transpose of matrix is:
1      3
2      4

-----
Process exited after 10.59 seconds with return value 0
Press any key to continue . . .

```